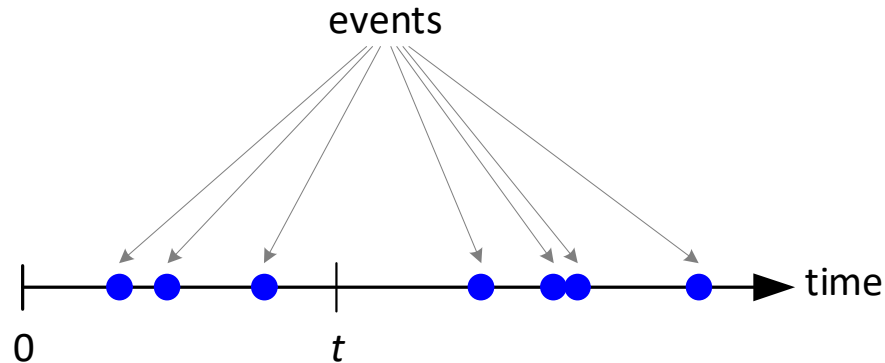


Poisson process

Poisson process

- Appears often in nature when we are observing the aggregate effect of a large number of individuals or particles operating independently.
- Models sequences of events.
- Simplifies analytical tractability.
- In computer networks, it is a good model for:
 - The sequence of times at which call requests arrive at a telephone switch
 - The sequence of times at which packets arrive at an Internet core router (no consensus about this)

Sequences of events

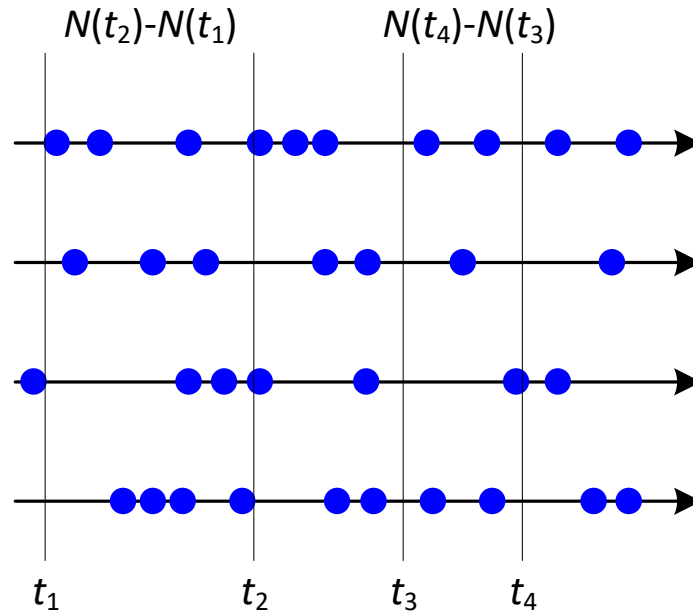


- A stochastic process $\{N(t), t \geq 0\}$ is a **counting process** if $N(t)$ represents the total number of events that have occurred up to time t .
- Examples:
 - births of children
 - people entering a building
 - goals scored by a particular soccer player

Sequences of events

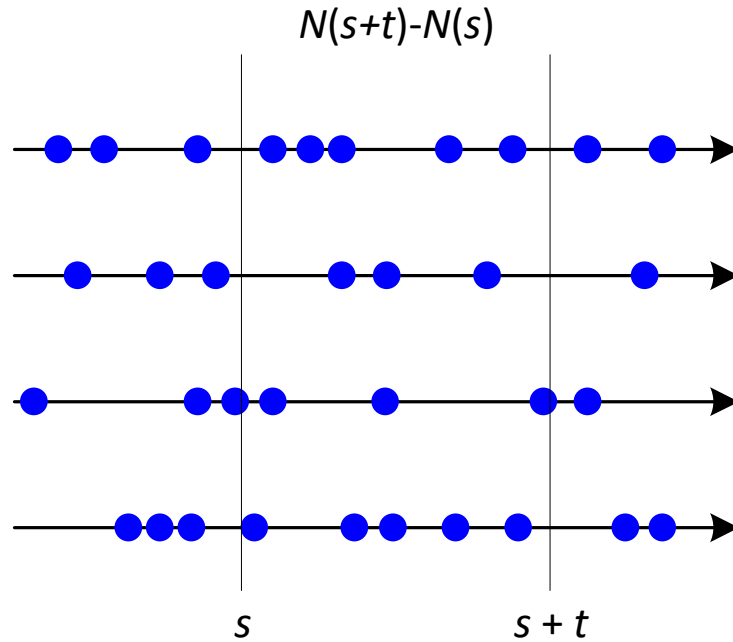
- A counting process has **independent increments** if the number of events in disjoint intervals is independent.
 - Birth of children has independent increments? No, birth rate depends on population, which increases with births
 - People entering a building has independent increments? Yes
 - Goals scored by a particular soccer player has independent increments? Maybe, depends on whether we believe in slumps!
- A counting process has **stationary increments** if the distribution of the number of events that occurs in any interval depends only on the length of the interval.

Independent increments



Number of events in $[t_1, t_2]$ independent of number of events in $[t_3, t_4]$

Stationary increments



Distribution of the number of events in $[s, s + t]$ independent of s ,
dependent only on t

*French Mathematician
(1781-1840)*

Definition 1 of Poisson process

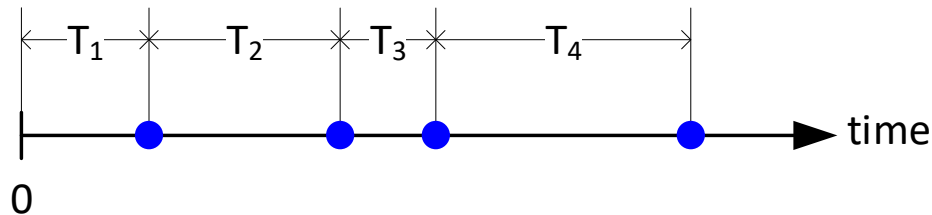
- A counting process is a Poisson process with rate λ , $\lambda > 0$, if
 - $N(0) = 0$;
 - The process has **independent increments**;
 - The number of events in an interval t is Poisson distributed with mean λt , i.e., for all $s, t \geq 0$

$$P(N(s+t) - N(s) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

- A Poisson process has **stationary increments** and mean $E[N(t)] = \lambda t$.

Definition 2 of Poisson process

- A Poisson process with rate λ is a counting process such that the interarrival times $T_1, T_2, T_3, \dots$ are exponential random variables with rate λ and $N(0) = 0$.
- Show that definition 1 implies definition 2



Exponential distribution:

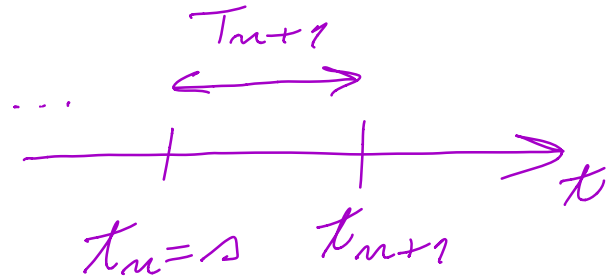
$$P(T > t) = e^{-\lambda t}, \quad t \geq 0$$

Poisson distribution:

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

$$\begin{aligned} T_1: \quad P(T_1 > t) &= P(N(t) = 0) \\ &= \frac{e^{-\lambda t} (\lambda t)^0}{0!} = e^{-\lambda t} \end{aligned}$$

$T_{n+1}, n \geq 0:$



$$P(T_{n+1} > t \mid \sum_{i=1}^n T_i = s) =$$

$\hookrightarrow$ time of n -th arrival

$$P(0 \text{ events } (s, s+t) \mid \sum_{i=1}^n T_i = s) =$$

$$= P(0 \text{ events } (s, s+t)) =$$

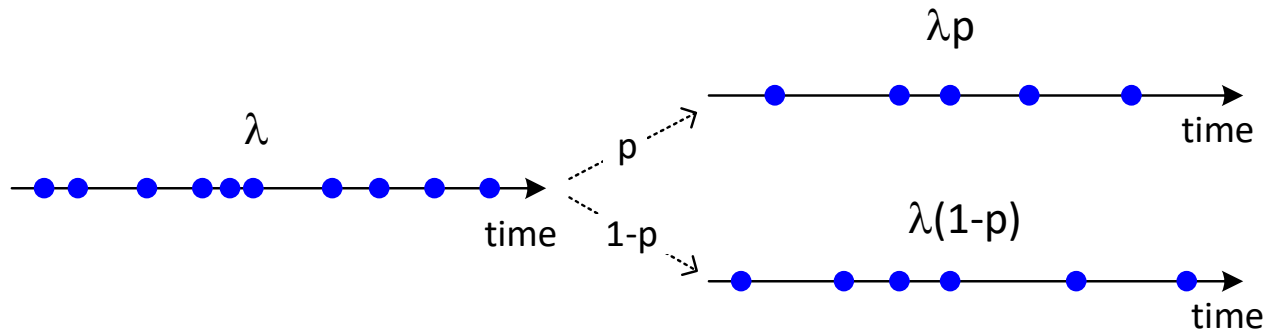
independent increments

$$= e^{-\lambda t}$$

stationary increments

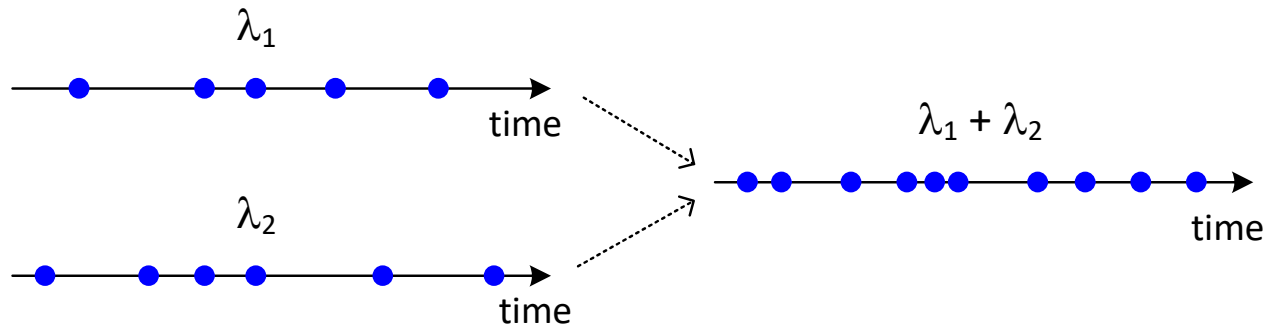
Poisson process - properties

- **Poisson splitting:** Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ . Each event is classified independently as a type 1 event with probability p or type 2 event with probability $1 - p$. $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are the number of events that have occurred in interval $[0, t]$. $N_1(t)$ and $N_2(t)$ are both Poisson processes with rates λp and $\lambda(1 - p)$.



Poisson process - properties

- **Poisson merging:** Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be independent Poisson processes with rates λ_1 and λ_2 . The process $N(t) = N_1(t) + N_2(t)$ is also a Poisson process with rate $\lambda = \lambda_1 + \lambda_2$.

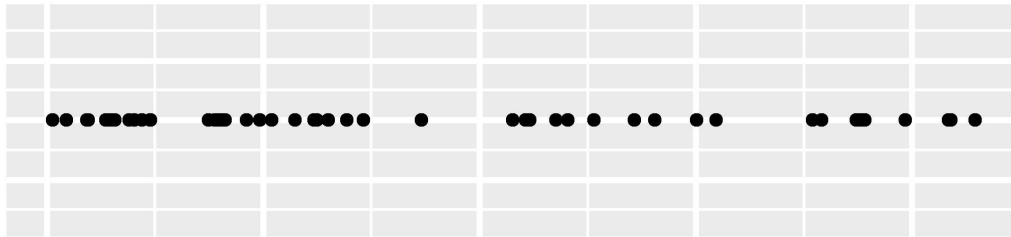


Poisson process – properties

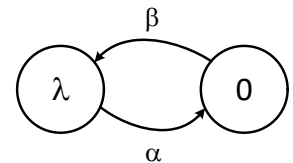
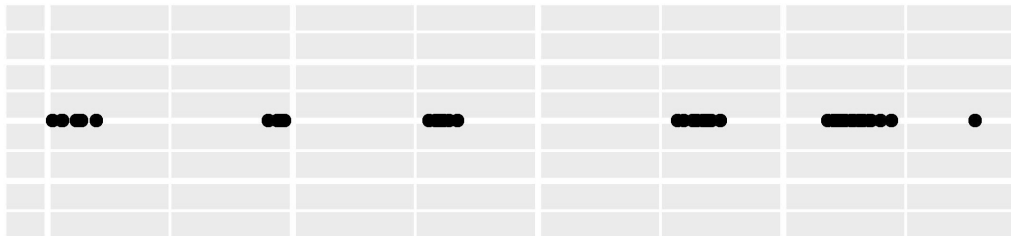
- **Uniformity**: Suppose that in a Poisson process exactly one event has occurred up to time t . Then the time at which the event has occurred is uniformly distributed in interval $[0, t]$, i.e. for $s < t$, $P\{T_1 < s \mid N(t) = 1\} = s/t$.
 - This result can be generalized in the following way: suppose that n events have occurred in interval $[0, t]$; the times $S_1, \dots, S_n$ at which the events have occurred, considered as non-ordered r.v. are distributed independently and uniformly distributed in interval $[0, t]$.
 - This is why the Poisson process is usually referred to as a process with **random arrivals**.

Poisson process - properties

Poisson



IPP



Poisson arrivals versus bursty arrivals (IPP)